

1. Distribución de los **label_3m = 1** por estado

```
positive_state_distribution <- sdf_sql(  
  sc,  
  "  
  WITH positives AS (  
  
    SELECT  
      party_id,  
      observation_date  
  
    FROM bu_frd.model_features_ab_final_2022_2025  
  
    WHERE label_3m = 1  
  ),  
  
  positive_state AS (  
  
    SELECT  
      f.party_id,  
      f.observation_date,  
      p.state_cde  
  
    FROM positives f  
  
    INNER JOIN bc3.padd p  
  
      ON f.party_id = p.party_id  
      AND p.data_date = f.observation_date  
      AND p.addr_seq_num = 1  
  
    WHERE p.data_date >= TO_DATE('2022-01-01')  
      AND p.data_date < TO_DATE('2026-01-01')  
  )  
  
  SELECT  
    COALESCE(state_cde, 'MISSING') AS state_cde,  
  
    COUNT(*) AS positive_parties,  
  
    COUNT(*) * 1.0 /  
    SUM(COUNT(*)) OVER () AS pct_of_positives  
  
  FROM positive_state  
  
  GROUP BY  
    COALESCE(state_cde, 'MISSING')
```

```
ORDER BY
  positive_parties DESC
"
) %>%
  collect()
```

positive_state_distribution

La suma de **positive_parties** debe volver a ser:

62,599

2. Más importante: balance Positive vs Negative por estado

Este es el que realmente responde al requirement que te dieron sobre:

balance between victims / non-victims by location

Aquí ya no filtramos **label_3m = 1**. Utilizamos toda la población analítica:

```
state_balance <- sdf_sql(
  sc,
  "
  WITH cohort_state AS (

    SELECT
      f.party_id,
      f.observation_date,
      f.label_3m,
      p.state_cde

    FROM bu_frd.model_features_ab_final_2022_2025 f

    INNER JOIN bc3.padd p

      ON f.party_id = p.party_id
      AND p.data_date = f.observation_date
      AND p.addr_seq_num = 1

    WHERE p.data_date >= TO_DATE('2022-01-01')
      AND p.data_date < TO_DATE('2026-01-01')
  ),

  state_counts AS (
```

```

SELECT
    COALESCE(state_cde, 'MISSING') AS state_cde,

    SUM(
        CASE WHEN label_3m = 1
            THEN 1 ELSE 0
        END
    ) AS positive_parties,

    SUM(
        CASE WHEN label_3m = 0
            THEN 1 ELSE 0
        END
    ) AS negative_parties,

    COUNT(*) AS total_parties

FROM cohort_state

GROUP BY
    COALESCE(state_cde, 'MISSING')
),

```

totals AS (

```

SELECT
    SUM(positive_parties) AS total_positive,
    SUM(negative_parties) AS total_negative,
    SUM(total_parties) AS total_population

FROM state_counts
)

```

```

SELECT
    s.state_cde,

    s.positive_parties,
    s.negative_parties,
    s.total_parties,

    s.positive_parties * 1.0 /
        t.total_positive AS pct_positive_distribution,

    s.negative_parties * 1.0 /
        t.total_negative AS pct_negative_distribution,

    s.total_parties * 1.0 /
        t.total_population AS pct_total_population,

```

```

CASE
  WHEN s.negative_parties > 0
  THEN
    (s.positive_parties * 1.0 / t.total_positive)
    /
    (s.negative_parties * 1.0 / t.total_negative)
  END AS representation_ratio,

  s.positive_parties * 1.0 /
  s.total_parties AS positive_rate_in_sample

FROM state_counts s

CROSS JOIN totals t

ORDER BY
  s.total_parties DESC
"
) %>%
collect()

state_balance

```

Este segundo query nos dará una tabla mucho más útil:

State	Positive Count	Negative Count	Total	% Positive s	% Negative s	Representation Ratio	Sample Positive Rate
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La columna más útil para revisar **balance** será:

$\text{RepresentationRatio}_s = \frac{\% \text{Positives}_s}{\% \text{Negatives}_s}$

Interpretación:

$$RR \approx 1$$

indica composición muy similar entre positives y negatives.

Si:

$$RR > 1$$

el estado está relativamente más representado entre positives.